

AI Context Project

Phase 1 – Week 3 Technical Documentation

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1. Introduction

The AI Context project focuses on developing an intelligent context and asset-search system as part of the AEV Platform. During the previous phases, the project established the backend environment and developed the initial Retrieval-Augmented Generation (RAG) pipeline for searching and retrieving relevant asset information.

Week 3 focuses on extending the AI Context service with a memory management layer and improving the prompt-template system. The memory component is designed to store useful user-specific context, while the prompt-template component provides structured and reusable instructions for generating AI responses.

The Week 3 implementation includes memory CRUD operations, user-memory isolation, memory categorization, memory expiry handling, privacy documentation, prompt template versioning, and authoring of three core templates.

2. Objective

The primary objective of Week 3 is to implement the first version of the memory management and prompt-template management functionality for the AI Context service.

The main objectives are:

- Implement Create, Read, Update, and Delete operations for user memory.
- Ensure that one user's memory cannot be accessed by another user.
- Support the four defined memory types:
 - Preference
 - Workflow
 - Role Context
 - Recent Topic
- Implement an expiry mechanism for unused memories.

- Document the privacy behaviour of stored memory.
 - Implement prompt template versioning.
 - Author the three required prompt templates.
 - Test memory isolation and expiry behaviour.
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- Document the implementation and testing results.

4. Memory API

The Memory API provides the basic operations required to manage user-specific memory within the AI Context service.

The API follows the standard CRUD model:

- **Create** – Store a new memory for a user.
- **Read** – Retrieve stored memory associated with a user.
- **Update** – Modify an existing memory.
- **Delete** – Remove an existing memory.

The memory API is intended to provide a structured interface for managing contextual information that can be used by the AI Context service.

Memory API Workflow

User Request → Memory API → User Identification → Memory Operation → Database

Memory Data

Each memory record should contain information required to identify the user, classify the memory, store the memory content, and maintain relevant timestamps.

[INSERT ACTUAL MEMORY MODEL / DATABASE SCHEMA SCREENSHOT HERE]

Figure 4.1 – Memory Data Model

5. User Memory Isolation

Memory isolation ensures that user-specific memory remains associated with the correct user.

The system must prevent one user from reading, modifying, or deleting another user's stored memory.

The general isolation workflow is:

User Request → User Identification → User-Specific Memory Query → Memory Result

Memory operations should include the appropriate user identifier when accessing stored memory.

Isolation Requirement

For a request made by User A:

- User A can access User A's memory.
- User A must not access User B's memory.
- User A must not modify User B's memory.
- User A must not delete User B's memory.

[INSERT ACTUAL ISOLATION CODE / TEST SCREENSHOT HERE]

Figure 5.1 – User Memory Isolation

6. Memory Types

The Week 3 memory system supports four memory categories.

Memory Type	Purpose
Preference	Stores user preferences that may be useful for future interactions
Workflow	Stores information related to a user's recurring workflow or working process
Role Context	Stores information relevant to the user's role or responsibilities
Recent Topic	Stores information about topics recently discussed by the user

Categorizing memories allows the system to distinguish between different types of contextual information.

Memory Classification

User Context → Memory Type → Memory Storage

[INSERT MEMORY TYPE IMPLEMENTATION SCREENSHOT HERE]

Figure 6.1 – Memory Type Implementation

7. Memory Expiry Job

The memory system includes an expiry mechanism for removing memories that have not been used for an extended period.

According to the Week 3 task specification, memories that have not been used for **90 days** are eligible for removal.

The expiry workflow can be represented as:

Stored Memory → Check Last Usage → 90 Days Without Use → Expiry → Memory Removal

The expiry process helps prevent outdated contextual information from remaining in the system indefinitely.

[INSERT ACTUAL EXPIRY JOB CODE / DRY-RUN SCREENSHOT HERE]

Figure 7.1 – Memory Expiry Job

8. Memory Privacy and Usage

User memory is intended to support contextual AI responses and should remain isolated to the user for whom it was stored.

Memory should not be treated as general training data for the AI model. The stored information is used as contextual information for the corresponding user's interactions.

The privacy behaviour of the memory system should therefore ensure:

- User memory remains associated with the correct user.
- Memory is not exposed to other users.
- Memory is used only for its intended contextual purpose.
- Outdated memory can be removed through the expiry mechanism.

[INSERT FINAL PRIVACY NOTE / IMPLEMENTATION EVIDENCE HERE]

9. Prompt Template Versioning

Prompt template versioning is used to maintain different versions of a prompt template when its instructions are modified.

Instead of replacing the previous template permanently, the system maintains an older version and creates a new version for the updated template.

Versioning Workflow

Template v1 → Template Updated → Template v2

The versioning system should allow the application to identify the currently active template while retaining previous versions for reference and controlled changes.

Example

Template	Version Status
Asset Summary v1	Previous
Asset Summary v2	Current

[INSERT ACTUAL TEMPLATE VERSIONING CODE / DATABASE SCREENSHOT HERE]

Figure 9.1 – Prompt Template Versioning

10. Prompt Template Authoring

Three prompt templates were defined for the AI Context service:

1. Asset Summary
2. Exposure Explanation
3. Remediation Plan

The templates provide structured instructions for processing retrieved asset information and generating consistent AI responses.

The templates follow the general structure:

Retrieved Context → Instructions → Output Requirements → AI Response

Each template is designed for a specific purpose within the asset-analysis workflow.

11. Template 1 – Asset Summary

The Asset Summary template is responsible for generating a concise and structured summary of retrieved asset information.

The template receives relevant asset context and instructs the AI model to summarize the available information without introducing unsupported information.

Purpose

The Asset Summary template is used to:

- Identify the relevant asset.
- Summarize its available information.
- Present important asset characteristics.
- Use only the retrieved context.

Prompt Structure

Context: Retrieved asset information.

Instruction: Generate a concise summary based on the supplied asset context.

Output Requirement: Return a structured summary supported by the provided information.

[PASTE ACTUAL FINAL ASSET SUMMARY PROMPT HERE]

[INSERT PROMPT IMPLEMENTATION SCREENSHOT HERE]

Figure 11.1 – Asset Summary Prompt Template

12. Template 2 – Exposure Explanation

The Exposure Explanation template is intended to explain the security exposure associated with a retrieved asset using the available asset context.

The response should remain grounded in the information supplied to the model.

Prompt Structure

Context: Relevant asset and security information.

Instruction: Explain the identified exposure using the provided context.

Output Requirement: Present the explanation clearly and avoid unsupported assumptions.

[PASTE ACTUAL FINAL EXPOSURE EXPLANATION PROMPT HERE]

[INSERT PROMPT IMPLEMENTATION SCREENSHOT HERE]

Figure 12.1 – Exposure Explanation Prompt Template

13. Template 3 – Remediation Plan

The Remediation Plan template is intended to generate a structured remediation approach based on the available asset and security context.

The generated recommendations should be based on the information provided to the model.

Prompt Structure

Context: Retrieved asset and exposure information.

Instruction: Generate a structured remediation plan based on the supplied context.

Output Requirement: Present actionable remediation steps supported by the available information.

[PASTE ACTUAL FINAL REMEDIATION PLAN PROMPT HERE]

[INSERT PROMPT IMPLEMENTATION SCREENSHOT HERE]

Figure 13.1 – Remediation Plan Prompt Template

17. Conclusion

Week 3 extends the AI Context service beyond the initial RAG pipeline by introducing user-specific memory management and an improved prompt-template system.

The planned memory functionality includes CRUD operations, user-memory isolation, four memory categories, and an expiry mechanism for unused memories. The prompt-template component introduces versioning and three specialized templates for asset summaries, exposure explanations, and remediation planning.

These components provide the foundation for maintaining contextual information and generating more consistent, structured responses within the AI Context service.

Further development can focus on integrating the memory system with the existing RAG workflow, refining prompt behaviour, and validating the system using broader test scenarios.